

Local convergence rates computed from adjacent refinement levels. The final row shows fitted rates with uncertainties from the log-log least-squares fit. Reynolds number 100.

cycle	# cells	# dofs	L2_velocity	rate	L2_pressure	rate	H1_velocity	rate
0	25	278	9.375e-03	–	1.248e-02	–	1.673e-01	–
1	100	1003	8.233e-04	3.51	2.678e-03	2.22	3.156e-02	2.41
2	400	3803	5.962e-05	3.79	6.530e-04	2.04	5.067e-03	2.64
3	1600	14803	4.334e-06	3.78	1.627e-04	2.01	9.016e-04	2.49
4	6400	58403	3.714e-07	3.54	4.065e-05	2.00	1.930e-04	2.22
fitted rate			–	3.682 ± 0.030	–	2.057 ± 0.025	–	2.46 ± 0.04

Table 1: Q2/Q1 elements

cycle	# cells	# dofs	L2_velocity	rate	L2_pressure	rate	H1_velocity	rate
0	25	633	1.886e-04	–	8.651e-04	–	6.613e-03	–
1	100	2363	8.138e-06	4.53	1.180e-04	2.87	5.976e-04	3.47
2	400	9123	3.287e-07	4.63	1.531e-05	2.95	5.392e-05	3.47
3	1600	35843	1.450e-08	4.50	1.944e-06	2.98	5.567e-06	3.28
4	6400	142083	7.671e-10	4.24	2.447e-07	2.99	6.495e-07	3.10
fitted rate			–	4.49 ± 0.04	–	2.950 ± 0.013	–	3.34 ± 0.05

Table 2: Q3/Q2 elements

cycle	# cells	# dofs	L2_velocity	rate	L2_pressure	rate	H1_velocity	rate
0	25	1138	4.935e-06	–	2.245e-05	–	2.553e-04	–
1	100	4323	9.466e-08	5.70	1.377e-06	4.03	1.038e-05	4.62
2	400	16843	1.868e-09	5.66	8.558e-08	4.01	4.688e-07	4.47
3	1600	66483	4.381e-11	5.41	5.341e-09	4.00	2.527e-08	4.21
4	6400	264163	1.224e-12	5.16	3.346e-10	4.00	1.508e-09	4.07
fitted rate			–	5.50 ± 0.07	–	4.008 ± 0.003	–	4.34 ± 0.07

Table 3: Q4/Q3 elements